
ME 310P: Thermo – Fluids laboratory

Course Code	ME 310P
Course Name	Thermo – Fluids laboratory
L-T-P-C	0-0-3-2
Prerequisites	ME 303 Heat Transfer and ME 210 Fluid Mechanics
Intended for	B.Tech.
Distribution	Core
Approval	8th Senate (Oct 2015)

Course Contents

- Uncertainty in measurements, curve fitting, introduction to the use of temperature measuring devices – thermometers, RTD, Thermocouples
- Determination of the calorific values of unknown fuels using the Bomb Calorimeter
- Evaluation of Energy efficiency of different stoves
- Parallel and series pump characteristics
- Finding the minor and major losses in pipes
- Flow visualization
- Determination of heat transfer coefficient during the natural convection
- Determination of heat transfer coefficient during the forced convection
- Shell and tube heat exchanger analysis under parallel and counter flow conditions
- Static pressure measurement on an aerofoil in wind tunnel
- Energy balance of 4 – S engine and exhaust gas analysis of the engine
- Vapour compression refrigeration test rig
- Determination of critical heat flux in pool boiling